

Co-LMLM Deletion Audit

What We Found Before Testing Identity Clustering

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Short version

We ran a three-state deletion audit on the public `lil-lab/CoLMLM-360M-FW` model and its released Wikipedia index. We used 1,500 examples from each of T-REx, PopQA, Google-RE, CounterFact, and ZsRE. Our detailed T-REx cohort contains 499 cases where the unmodified model is correct and retrieves an answer-mentioning span before the answer is visible.

Deleting only that retrieved row leaves 39.1% of the T-REx answers intact. Deleting its cosine neighborhood leaves 31.1%. A filter supplied with the gold answer and aliases reduces survival to 6.8%, and combining that filter with geometry reduces it to 5.8%. This last result is a useful efficacy upper baseline, but it is not a deployable deletion rule: it uses the ground-truth answer at inference time, can remove unrelated facts that share the same answer, and guarantees that our answer-mention trace check is zero.

The same combined policy leaves between 0.3% and 5.8% intact across the five datasets. On T-REx, 23.2% of the cohort remains answerable when factual retrieval is disabled, and retrieval changes a correct retrieval-disabled answer into a wrong answer in 18.4% of cases. A synthetic answer-bearing row placed just outside the cosine boundary also restores many answers. Our main conclusion is therefore narrower than “semantic deletion works”: a database row, a source, a cosine neighborhood, and an answer string are different from a proposition identity. This note is the audit before identity clustering; identity clustering itself is not implemented here.

1 Scope and relation to the Co-LMLM paper

Co-LMLMexternalizes free-form factual spans into a continuous-key, text-value index and retrieves them from hidden-state queries [1]. The paper reports aggregate forgetting on TOFU [2]. Our run complements that experiment; it does not reproduce it. The TOFU experiment fine-tunes on synthetic biographies, adds those biographies to the knowledge base, and removes entries for a designated forget set. We instead audit naturally occurring retrieval paths in the released Wikipedia index.

The question here is: after a deletion request, does the answer still arrive through another memory entry, or can the model and prompt produce it without factual retrieval? This separates failures of the deletion set from answers that remain available through the prompt, model parameters, or general language patterns.

This is also the intended step before identity-key research. We first test several current meanings of “delete this fact”—one row, one source, a cosine neighborhood, or a matching answer string. We can then compare those baselines with a persistent proposition identity.

2 Audit protocol

Our design adapts the causal audit used for relational LLMs [3]. For each fact, the prompt, model, decoding, and parameters stay fixed while memory access changes:

State	Intervention
FULL	Memory unchanged; retrieval enabled.
DEL-ON	Target deletion set hidden; retrieval enabled.
DEL-OFF	Target deletion set hidden; factual retrieval disabled.

Let $C_s(f)$ indicate whether the output in state s contains the normalized gold answer or an accepted alias as a complete phrase. Conditional on FULL correctness, we report

$$L(f) = C_{\text{DEL-OFF}}(f), \quad \text{retrieval-disabled answerability,} \quad (1)$$

$$R(f) = C_{\text{DEL-ON}}(f)(1 - C_{\text{DEL-OFF}}(f)), \quad \text{retrieval-mediated correctness,} \quad (2)$$

$$I(f) = (1 - C_{\text{DEL-ON}}(f))C_{\text{DEL-OFF}}(f), \quad \text{retrieval interference,} \quad (3)$$

$$S(f) = C_{\text{DEL-ON}}(f), \quad \text{post-deletion survival.} \quad (4)$$

The correctness rule is directional: the complete normalized answer or alias must appear in the model output. A shorter prediction contained inside the answer does not count. For example, “Black” is not accepted for “Blackpool.”

2.1 Cohort construction

We audit the first retrieval event selected by the backend that satisfies all of the following: it returned an entry ID, the entry passes our normalized answer-mention heuristic, and the answer was not already visible before that retrieval event. The heuristic is not a proposition verifier. It checks whether the retrieved text contains the normalized answer or alias; it also accepts a retrieved span of at least three characters contained inside the answer. In the T-REx cohort, 570 entries pass by direct answer mention and two pass only by this reverse check.

Table 1: Construction of the auditable cohorts. “Pre-answer mention” means that the selected retrieved span passed the answer-mention heuristic before the answer was visible. The primary cohort also requires a correct FULL answer.

Dataset	Prompts	Selected entry	Pre-answer mention	Primary
T-REx	1,500	1,095	572	499
PopQA	1,500	1,438	618	468
Google-RE	1,500	1,475	787	759
CounterFact	1,500	1,102	439	358
ZsRE	1,500	1,042	535	337

For T-REx, the 928 exclusions are 405 prompts with no selected entry, 509 whose selected span does not pass the answer-mention heuristic, and 14 where the answer is already visible before the answer-mentioning event. The unmodified model is correct on 746 of all 1,500 T-REx prompts under our scoring rule. The 499-case primary cohort is 33.3% of the full slice.

2.2 Run configuration

All five runs use greedy decoding with at most 12 new tokens, retrieval similarity threshold 0.7, the public `lil-lab/CoMLM-360M-FW` checkpoint, and the released approximate Wikipedia index of about 240M entries. They do not use the 2.2B-entry Wikipedia–FineWeb-Edu index [1]. T-REx and PopQA use stable seeded slices; Google-RE is balanced across place of birth and place of death; CounterFact and ZsRE use the first 1,500 eligible records. The T-REx sample, 12-token limit, and containment metric differ from the paper’s published benchmark protocol, so the FULL score above is not a replication of the paper’s T-REx result.

Deletion is implemented through reversible search-time filtering rather than physical index rewriting. Every successful retrieval event was aligned to its generated `<FACT>` block, and the geometric closure was centered on the same audited event used to select the deletion target.

3 Results

3.1 Five-dataset summary

The standard policy combines geometric deletion at $\rho = 0.85$ with the gold-answer string filter. Table 2 reports the paired outcomes on each primary cohort.

Table 2: Three-state results under geometric plus gold-answer string filtering. All rates are conditional on FULL correctness and pre-answer answer-mention retrieval.

Dataset	Primary	Survival S	Disabled L	Mediated R	Interference I
T-REx	499	5.8%	23.2%	1.0%	18.4%
PopQA	468	2.4%	8.1%	1.3%	7.1%
Google-RE	759	0.3%	0.3%	0.0%	0.0%
CounterFact	358	4.7%	20.4%	0.3%	15.9%
ZsRE	337	0.6%	0.9%	0.3%	0.6%

The size of the retrieval-disabled channel varies strongly by prompt format: it is substantial on T-REx and CounterFact, smaller on PopQA, and nearly absent on Google-RE and ZsRE. The deletion-policy result is more consistent: combined-policy survival is at most 5.8% on every dataset. Because the policy receives the gold answer, this consistency measures end-to-end efficacy of an oracle string filter, not successful proposition resolution.

3.2 T-REx channel decomposition

Table 3: Paired T-REx outcomes under geometric plus gold-answer string filtering at $\rho = 0.85$, restricted to the 499-case primary cohort.

Outcome	Count	Rate
Wrong in both DEL-ON and DEL-OFF	378	75.8%
Correct only in DEL-ON: retrieval-mediated R	5	1.0%
Correct only in DEL-OFF: retrieval interference I	92	18.4%
Correct in both post-deletion states	24	4.8%
Any DEL-ON correctness: survival S	29	5.8%
Any DEL-OFF correctness: L	116	23.2%

The standard policy leaves 29 of 499 answers intact (5.8%, Wilson 95% interval [4.1, 8.2]). Only five are correct uniquely with retrieval. By contrast, 116 are correct without factual retrieval. In 92 cases, retrieval changes a correct retrieval-disabled answer into a wrong answer.

All 29 DEL-ON-correct traces lack a retained answer mention. This is expected by construction: the same answer-mention rule is used both to filter candidates and to inspect the trace. It is therefore a sanity check on the implementation, not independent evidence that all semantic support was removed.

The DEL-OFF result is similar under the stricter control that forbids the retrieval token: 110 of 499 are correct (22.0%), compared with 116 (23.2%) under null retrieval. The paired difference is 1.2 percentage points ($p = 0.362$). We call this retrieval-disabled answerability rather than pure weight memorization because the prompt and general language patterns remain available.

3.3 Deletion policy and observed scope

We compare six end-to-end policies. Selected-entry deletion hides the audited row. Provenance deletion also hides entries from its source. Geometric deletion hides keys with cosine similarity at least 0.85. Gold-answer string filtering receives the target answer and aliases at inference time and hides matching candidates. The combined and hybrid policies take unions of these rules.

Table 4: T-REx policy comparison on the 499-case primary cohort. Scope statistics use all 572 pre-answer answer-mention targets. “Observed excluded IDs” is the union of the saved manifest and candidate IDs encountered and filtered during generation; it is a lower bound on total policy scope. Provenance scope is source-wide and cannot be counted from the saved trace. † Zero answer mentions are guaranteed by the gold-answer filter.

Policy	Survival	Efficacy	Mention in survivor	Mean	Median	Max.
Selected entry	39.1%	60.9%	178/195	1.0	1	1
Provenance	39.7%	60.3%	185/198	—	—	—
Geometric, $\rho = 0.85$	31.1%	68.9%	139/155	2.6	1	183
Gold-answer string filter	6.8%	93.2%	0/34†	79.8	23	2,120
Geometric + gold-answer	5.8%	94.2%	0/29†	86.5	24	2,835
Hybrid	5.8%	94.2%	0/29†	88.4	25	2,835

Deleting one observed row is not the same as deleting the fact: 178 of the 195 selected-entry survivors retrieve another answer-mentioning span. Geometric closure removes more, but 139 of its

155 survivors still retrieve an answer mention. The same record–fact mismatch has been identified as a structural problem for selective forgetting in RAG systems [4].

The gold-answer filter produces the large efficacy gain. There are 124 cases that survive geometric deletion but not gold-answer filtering, versus three in the reverse direction ($p < 5 \times 10^{-33}$). This is an oracle upper baseline, not a semantic policy. Different subjects and relations can share the same answer, so a string rule can remove lawful and unrelated facts.

The saved top-500 value manifests contain a mean of 60.0 entries per target (median 22, maximum 494). During generation, the filter also applies to newly encountered candidates. Taking the union of saved and dynamically observed IDs raises the mean to 79.8, the median to 23, and the maximum to 2,120. These are still observed lower bounds, not the number of matching entries in the whole 240M-entry index.

There is also an approximate-search confound. Selected-entry and geometric policies search at shallow, target-dependent depths, while value and provenance filtering search about 4,097 candidates at a time. On a compressed approximate index, changing search depth can change the apparent top result. The table therefore compares complete inference policies; it does not isolate deletion-set membership alone. A fixed-depth validation subset is needed for that stronger causal comparison.

3.4 Geometric boundary stress test

For each T-REx target, this test inserts one synthetic answer-bearing entry at similarity $\rho - \epsilon$, just outside the geometric deletion radius $\rho = 0.85$. The baseline is geometric deletion alone, whose survival rate is 31.1%.

Table 5: Synthetic verbatim row placed outside the geometric radius. Attack gain is the fraction changed from wrong under baseline geometric deletion to correct after insertion.

ϵ	Synthetic row selected	Post-attack correct	Attack gain
0.01	87.2%	59.3%	36.1%
0.02	77.2%	58.3%	33.3%
0.05	56.3%	50.9%	24.0%

An explicit answer value immediately outside the radius is often selected and restores the answer. The same qualitative effect appears in all five datasets. This shows that the cosine threshold is a routing boundary, not a semantic deletion guarantee. Hyphenated, letter-spaced, and prefix-cued values add almost no new correct answers on T-REx (attack gain at most 0.2%). This is a synthetic verbatim boundary test, not evidence about natural paraphrases or general obfuscation.

3.5 Secondary diagnostics

Representation probe. The probe is a five-fold closed-set ridge-ranking test, not an open-answer decoder. It chooses among 315 candidate answers using event-0 FULL query vectors. It selects the accepted answer for 125 of 572 facts (21.9%), compared with 41 of 572 (7.2%) for the most common answer. Behavioral DEL-OFF correctness is 130 of 572 (22.7%), but the two sets overlap weakly (Jaccard 19.2%). For 13 cohort cases, event 0 is not the later audited answer-mention event, so this remains a descriptive diagnostic.

Radius sweep and collateral. On T-REx, geometric efficacy rises from 61.7% at $\rho = 0.95$ to 76.8% at $\rho = 0.70$. Measured collateral is zero through $\rho = 0.75$ and 0.22% at $\rho = 0.70$. We do

not treat this as a headline result. In 450 of 499 target neighborhoods, the requested cosine ball contained too few comparison facts and was filled with facts outside the ball; only 552 of 2,607 evaluated links were inside it. The approximate search-depth issue also applies to this sweep.

4 What we think is worth testing next

Let each memory entry be $e_i = (k_i, v_i, s_i)$, containing a continuous key, text value, and source metadata. The next useful baseline is a persistent identity assignment z_i such that

$$z_i = z_j \iff e_i \text{ and } e_j \text{ express the same deletable proposition,}$$

under a clear deletion policy. A request for identity z should hide every entry assigned to z , while retaining entries that merely share an answer, topic, or source.

For example, “X was born in Paris” and “Y was born in Paris” share an answer but express different propositions. “X’s birthplace is Paris” and “X was born in the French capital” may express the same proposition. Identity therefore needs more than unconstrained clustering of key vectors.

4.1 A minimal identity-aware baseline

1. Generate high-recall candidate pairs using key neighbors, entities, values, and source links.
2. Label pairs as *same proposition*, *same answer only*, *related*, or *unrelated*. Shared-answer examples are important hard negatives.
3. Score proposition equivalence using text, entity and relation consistency, provenance, and key similarity. Form constrained clusters and store an inspectable identity ID.
4. Delete by identity and rerun the same three-state audit with fixed search depth, cohort, and decoding.

The key comparison is whether identity deletion approaches the oracle filter’s efficacy while preserving unrelated entries that share its answer. We should report cluster precision and recall, cluster size, latency, survival, retrieval-disabled answerability, retrieval interference, and collateral on separate hard-negative sets for shared answer, entity, relation, source, and key neighborhood. Splits should be identity-disjoint.

Calibrated abstention is also worth testing. The interference rates show that accepting the next available candidate can turn a correct retrieval-disabled answer into a wrong answer. Abstention and identity resolution address different problems and can be evaluated separately.

5 What this note does not claim

- The five main results are conditional on FULL-correct cases with a pre-answer answer-mention retrieval. They do not measure all knowledge in the model or index.
- The answer-mention heuristic does not verify the full subject–relation–object proposition.
- Gold-answer filtering uses the target answer and aliases at inference time. It is an efficacy upper baseline, not a deployable deletion policy.
- DEL-OFF removes factual retrieval but does not separate model parameters from information in the prompt.

- Search-time filtering tests behavioral deletion, not physical erasure, replicas, caches, or later index rebuilds.
- Approximate retrieval and policy-dependent search depth limit causal comparisons between deletion sets.
- These 1,500-example slices and our decoding and scoring rules are not direct replications of the paper’s benchmark or TOFU experiments.
- Identity clustering, modified pretraining, and learned abstention have not yet been implemented.

6 Our overall read

The results are positive for the Co-LMLM direction: the external memory is editable, inspectable, and clearly affects behavior. The open problem is the unit of deletion. Selected-row and cosine policies miss many alternative answer-bearing spans. Gold-answer filtering closes most observed paths, but it is an oracle string rule and can be too broad. The audit uses event-aligned retrieval traces, paired interventions, controls, five-dataset coverage, and concrete baselines for the next step: proposition-level identity keys with measured efficacy and collateral damage.

Statistical note. Intervals are two-sided 95% Wilson intervals. Paired comparisons use two-sided exact McNemar tests; exploratory tests are not adjusted for multiplicity.

References

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